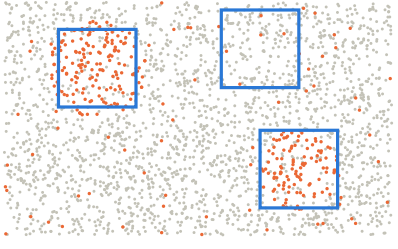
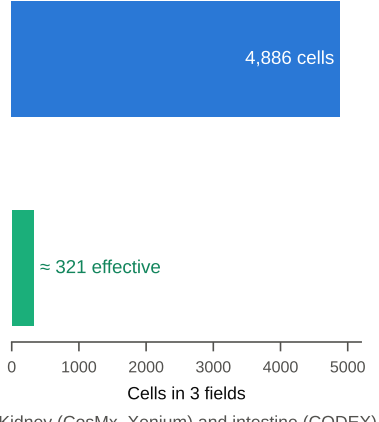
1 A few fields of view



Cells within a field are correlated:
fields, not cells, are the sampling units

2 Design effect ≈ 15–60



Kidney (CosMx, Xenium) and intestine (CODEX)

3 Consequences and remedy

Cells as replicates: 87–94% false positives
Cell-level intervals cover 20–37%

Individuals as units: 3–6% false positives
Field-model intervals cover 91–94%

fovplan: pilot fields → number of fields for a target precision (≈95% within target with the 90% assurance plan)